

Block Coding with Sphero

Piter Garcia

March 20, 2026

Packet Use

This printable lesson packet is generated from the paired Markdown guide. The Markdown version remains the clickable, neurodivergent-friendly working copy; this PDF carries the same lesson substance in a formal handout format. Evidence claims in this packet are based on transcript-derived lesson notes and the cleaned transcript files in this workspace.

Quick Scan

- **Grade:** Grade 3
- **Grade Hub:** Notes [../grade-3-Notes.md]
- **Schedule Reference:** Spring2026_TeachingSchedule_Derek.md [../school/Spring2026_TeachingSchedule_Ignite/grade-3/Spring2026_TeachingSchedule_Derek.md]
- **Workbook Sheet:** 3rd in Teaching_Placement-IGNITE Curriculum_ GCSD25-26.xlsx
- **Lesson Focus:** Students pair robots, create movement programs, use loops for repeated motion, and debug route accuracy through floor-based challenges.
- **CS Alignment Strength:** Strong CS lesson
- **LaTeX Source:** block-coding-with-sphero.tex [./block-coding-with-sphero.tex]
- **Bib File:** block-coding-with-sphero.bib [./block-coding-with-sphero.bib]
- **Artifacts Folder:** artifacts/ [./artifacts/]

Lesson Identity

- **Likely Class Blocks:** Day 3 – 9:50-10:40 – Lamanaco, Day 4 – 9:50-10:40 – Lallucci, Day 5 – 9:50-10:40 – Regelsberger, Day 5 – 2:15-3:05 – Tandoi
- **Main Lesson Family:** Block Coding with Sphero
- **Primary Evidence Base:** Transcript-derived notes plus source transcripts from this teaching-placement workspace.
- **Primary External Source Type:** Official curriculum/provider materials.

What We Are Teaching

- pairing and aiming a robot
- sequencing movement blocks into a route
- using loops or repeated structures to shorten code
- debugging distance, direction, and timing

CS Standards Alignment

- **1B-AP-10:** Create programs that include sequences, events, loops, and conditionals. Why it fits here: Students build working movement programs with repeated structures.
- **1B-AP-11:** Decompose problems into smaller, manageable subproblems. Why it fits here: Students break the route into single moves before combining them.
- **1B-AP-15:** Test and debug a program or algorithm to ensure it runs as intended. Why it fits here: Students repeatedly test, adjust, and rerun their robot code.

NYS / Local Curriculum Alignment

- Aligned to the local Grade 3 IGNITE robotics sequence in the workbook and schedule.
- Supports New York-facing computational thinking and collaborative problem solving through route planning, testing, and revision.
- This is a core CS packet because the content centers on algorithm design, repeated patterns, and debugging.

Lesson Content and Concepts

- sequence
- loop
- debugging
- robot control
- decomposition

Evidence / Proof of Instruction

- **Transcript-derived note:** 260227 Grade 3 Lamanaco Sphero Loops [../transcript-derived-lessons/260227-grade-3-lamanaco-sphero-loops.md] The lesson note explicitly names pairing, loops, square movement, aiming, and safety.

- **Transcript-derived note:** 260227 Grade 3 Lamanaco Block Coding [../transcript-derived-lessons/260227-grade-3-lamanaco-block-coding.md] The additional note shows the same lesson family emphasized core block coding rather than one isolated challenge.
- **Transcript-derived note:** 260306 Grade 3 Lamanaco Square Maze [../transcript-derived-lessons/260306-grade-3-lamanaco-square-maze.md] Later transcripts show students applying the same logic to more precise route and maze navigation tasks.
- **Source transcript:** 260227 Grade 3 Lamanaco Sphero Loops [../transcripts/260227-grade-3-lamanaco-sphero-loops.md] The source transcript records teacher language around loops, pairing, and safety while students work in robot pairs.
- **Likely student proof:** Working square paths, looped code blocks, debug revisions, and safe partner management are direct evidence of instruction.

Assessment Evidence

- Student pairs the correct robot and keeps it under control.
- Student writes a movement sequence that completes the task or gets close enough to debug.
- Student can explain one fix they made after testing.

UDL and Inclusion Supports

Multiple Representation

- Model the route on the floor and on screen before coding.
- Use arrows, distance markers, and turn visuals.
- Show one buggy program and one corrected version side by side.

Multiple Action / Expression

- Allow coding by dragging blocks, pointing to a route card, or verbally sequencing for a partner.
- Use clearly assigned pair roles: coder, launcher, observer, recorder.
- Give students a printed planning strip before the digital build.

Multiple Engagement

- Use robot motion as immediate feedback for effort and revision.
- Keep challenges short enough for multiple retry cycles.
- Frame debugging as the normal path to success.

How We Are Already Making It Inclusive

- Transcript evidence shows teacher modeling before student handling of the robots.
- Partner structures were already present, which supports participation for students who need shared problem solving.
- The lesson used physical movement and visible outcomes, which lowers abstraction load.

How to Improve Inclusion Next Time

- Add route mats with start/end icons and numbered steps.
- Create a quiet-observer role for students who need to participate without launching every round.
- Build a common error wall with examples like overshoot, wrong angle, and unpaired robot.
- Store route photos and code screenshots in **artifacts/** as proof.

Materials and Setup

- Sphero robots and chargers
- Student devices
- Floor route or taped path
- Planning strips or dry-erase route cards

Teacher Moves / Script / Routines

- Model pairing and aiming before any free handling begins.
- Have students predict the route before launch.
- Stop midway to compare a long program and a looped version.
- Close with a debug story: what failed and what changed.

Common Errors and Debugging

- Students may skip pairing steps and blame the code.
- Distance and angle errors can look like conceptual failure when they are calibration issues.
- Partner imbalance can leave one student passive without role structure.

Extensions and Simplifications

- Extension: compare two possible solutions and justify which is more efficient.
- Simplification: reduce the task to one turn and one forward move repeated.
- Extension: add an event or sensor-based trigger after the movement pattern works.

Related Local Materials

- Block Coding with Bees materials [[../lessons/23-Block-Coding-with-Bees/23-Block-Coding-with-Bees-Notes.md](#)]

Artifacts Checklist

- Compiled lesson PDF in **artifacts/**
- At least one screenshot, student example, or setup photo
- One short assessment or observation record
- Updated standards/source bibliography once the **.bib** file is revised
- Any teacher reflection or revision notes after reteaching

Selected Official Sources

Selected official sources that informed this packet are listed below. ocite*

References